

Prepare a lot of questions and their answers based on these topics. this is for my preparation in accenture internal training exam. help me prepare

Absolutely. I went through the uploaded Accenture training syllabus. It covers **AWS, SQL, Python, Big Data, AI/ML/DL/GenAI, Data Warehousing, Data Integration/Talend, PySpark, Synthetic Data, Databricks, Agentic AI, Git/GitHub, CI/CD, Docker, Kubernetes and Microservices.**

The **PySpark and Databricks sections are particularly extensive**, including RDDs, DataFrames, Spark architecture, Delta Lake, Lakehouse, Vector Search, RAG, Mosaic AI Agent Framework and Unity Catalog tool functions.

Below is a large **exam-oriented Q&A bank**. The uploaded document gives the topics, while the technical answers below use standard technical knowledge to build preparation material around those topics.

Accenture Internal Training Exam Question Bank

SECTION 1: AWS CLOUD FUNDAMENTALS

1. What is cloud computing?

Answer: Cloud computing is the delivery of computing resources such as servers, storage, databases, networking and software over the internet on demand.

2. What are the major advantages of cloud computing?

Answer: Scalability, elasticity, pay-as-you-go pricing, high availability, reduced infrastructure management and global accessibility.

3. What is AWS?

Answer: Amazon Web Services is a cloud platform providing compute, storage, database, networking, security, analytics and many other services.

4. What is the difference between scalability and elasticity?

Answer:

- **Scalability:** Ability to increase or decrease resources as requirements change.
- **Elasticity:** Automatic adjustment of resources according to demand.

5. What is the AWS Shared Responsibility Model?

Answer: AWS is responsible for security **of** the cloud infrastructure, while customers are responsible for security **in** the cloud, depending on the service.

6. What is an AWS Region?

Answer: A geographical area containing multiple AWS Availability Zones.

7. What is an Availability Zone?

Answer: A physically separate data-center location within an AWS Region designed to provide isolation and high availability.

8. Why deploy an application across multiple Availability Zones?

Answer: To improve availability and resilience if one Availability Zone becomes unavailable.

9. What is IAM?

Answer: AWS Identity and Access Management controls authentication and authorization for AWS resources.

10. What is the principle of least privilege?

Answer: Grant users and services only the permissions they actually require.

SECTION 2: AWS COMPUTE

The syllabus explicitly includes AWS Compute Services.

11. What is Amazon EC2?

Answer: Elastic Compute Cloud provides resizable virtual servers in AWS.

12. What is an EC2 instance?

Answer: A virtual machine running in AWS.

13. What is an AMI?

Answer: An Amazon Machine Image is a template containing information required to launch an EC2 instance.

14. What is Auto Scaling?

Answer: Automatically adding or removing compute resources according to demand.

15. What is a load balancer?

Answer: A service that distributes incoming traffic across multiple backend resources.

16. What is AWS Lambda?

Answer: A serverless compute service that executes code in response to events without requiring users to manage servers.

17. What is serverless computing?

Answer: A model where the cloud provider manages the underlying servers and infrastructure while users focus primarily on application code.

18. When would you choose Lambda instead of EC2?

Answer: For event-driven workloads, short-running functions and workloads where managing servers is unnecessary.

19. What is ECS?

Answer: Amazon Elastic Container Service is a managed container orchestration service.

20. What is EKS?

Answer: Amazon Elastic Kubernetes Service is AWS's managed Kubernetes service.

21. What is the difference between EC2 and Lambda?

Answer: EC2 provides persistent virtual machines that you manage, while Lambda runs functions without requiring server management.

SECTION 3: AWS STORAGE AND DATABASES

The syllabus contains two dedicated storage/database modules.

22. What is Amazon S3?

Answer: Amazon Simple Storage Service is an object storage service.

23. What is an S3 bucket?

Answer: A logical container used to store objects in S3.

24. What is an S3 object?

Answer: A file and its associated metadata stored in an S3 bucket.

25. Is S3 block storage or object storage?

Answer: Object storage.

26. What is EBS?

Answer: Elastic Block Store provides persistent block storage for EC2 instances.

27. S3 vs EBS?

Answer:

- S3: Object storage.
- EBS: Block storage attached to EC2.

28. What is Amazon RDS?

Answer: A managed relational database service supporting database engines such as PostgreSQL, MySQL and others.

29. Why use RDS instead of installing a database manually on EC2?

Answer: AWS handles many administrative tasks such as backups, patching and infrastructure management.

30. What is Amazon DynamoDB?

Answer: A managed NoSQL database service designed for low-latency, highly scalable applications.

31. SQL database vs NoSQL database?

Answer: SQL databases generally use structured relational schemas, while NoSQL databases support flexible data models such as key-value, document or wide-column models.

SECTION 4: AWS NETWORKING

Networking is explicitly included in the syllabus.

32. What is a VPC?

Answer: Virtual Private Cloud is an isolated virtual network within AWS.

33. What is a subnet?

Answer: A subdivision of an IP network inside a VPC.

34. Public subnet vs private subnet?

Answer: A public subnet has a route to an Internet Gateway, while a private subnet does not directly route internet traffic through an Internet Gateway.

35. What is an Internet Gateway?

Answer: A VPC component that enables communication between resources in a VPC and the internet.

36. What is a NAT Gateway?

Answer: It allows resources in private subnets to initiate outbound internet connections without accepting unsolicited inbound internet connections.

37. What is a Security Group?

Answer: A stateful virtual firewall associated with resources such as EC2 instances.

38. What is a Network ACL?

Answer: A subnet-level network traffic control mechanism.

39. Security Group vs NACL?

Answer:

- Security Group: Resource-level, stateful.
- NACL: Subnet-level, stateless.

40. What is DNS?

Answer: Domain Name System translates domain names into IP addresses.

SECTION 5: SQL

SQL is explicitly listed as a core module.

41. What is SQL?

Answer: Structured Query Language is used to interact with relational databases.

42. What is a primary key?

Answer: A column or combination of columns that uniquely identifies each row.

43. What is a foreign key?

Answer: A column that references a primary key or unique key in another table.

44. WHERE vs HAVING?

Answer:

- WHERE filters rows before aggregation.
- HAVING filters groups after aggregation.

45. What does GROUP BY do?

Answer: Groups rows having the same values for specified columns.

46. What does ORDER BY do?

Answer: Sorts query results.

47. What does DISTINCT do?

Answer: Removes duplicate rows from the result.

48. What is an INNER JOIN?

Answer: Returns rows where matching values exist in both tables.

49. What is a LEFT JOIN?

Answer: Returns all rows from the left table and matching rows from the right table.

50. What is a RIGHT JOIN?

Answer: Returns all rows from the right table and matching rows from the left table.

51. What is a FULL OUTER JOIN?

Answer: Returns matching rows plus unmatched rows from both tables.

52. What is normalization?

Answer: Organizing relational data to reduce redundancy and improve data integrity.

53. What is denormalization?

Answer: Intentionally introducing redundancy to improve query performance or simplify access.

54. DELETE vs TRUNCATE?

Answer:

- DELETE removes rows and can use a condition.
- TRUNCATE removes all rows from a table and is generally more efficient.

55. DROP vs TRUNCATE?

Answer:

- DROP removes the table itself.
- TRUNCATE removes the data while retaining the table structure.

56. What is an index?

Answer: A database structure used to speed up data retrieval.

57. What is a view?

Answer: A virtual table defined by a SQL query.

58. What is a subquery?

Answer: A query nested inside another SQL query.

59. What is a window function?

Answer: A function that performs calculations across related rows without collapsing them into a single row per group.

60. Give examples of window functions.

Answer: ROW_NUMBER(), RANK(), DENSE_RANK(), LEAD(), LAG() and aggregate functions used with OVER().

SECTION 6: PYTHON

The syllabus covers Python, EDA, data processing/visualization and Python with SQL databases.

61. List vs tuple?

Answer: Lists are mutable, while tuples are immutable.

62. List vs set?

Answer: Lists maintain ordered collections and allow duplicates. Sets contain unique elements and are optimized for membership operations.

63. What is a dictionary?

Answer: A mutable key-value data structure.

64. What is a Python function?

Answer: A reusable block of code that performs a specific operation.

65. What is a lambda function?

Answer: An anonymous function usually written as a single expression.

66. What is list comprehension?

Answer: A concise syntax for creating lists from iterable data.

67. What is exception handling?

Answer: Handling runtime errors using mechanisms such as try, except, else and finally.

68. What is NumPy?

Answer: A Python library primarily used for numerical computing and array operations.

69. What is Pandas?

Answer: A Python library for data manipulation and analysis.

70. What is a DataFrame?

Answer: A two-dimensional tabular data structure with rows and columns.

71. What is EDA?

Answer: Exploratory Data Analysis is the process of examining and understanding datasets using statistics and visualizations.

72. What is missing-value handling?

Answer: Identifying missing data and deciding whether to remove, replace or otherwise handle it.

73. What is data visualization?

Answer: Representing data graphically to identify patterns, trends and relationships.

SECTION 7: BIG DATA

The syllabus specifically covers Big Data, Data Lakes and NoSQL.

74. What is Big Data?

Answer: Extremely large, complex or rapidly generated datasets that require specialized technologies for storage and processing.

75. What are the traditional 5 Vs of Big Data?

Answer: Volume, Velocity, Variety, Veracity and Value.

76. What is Volume?

Answer: The amount of data.

77. What is Velocity?

Answer: The speed at which data is generated, transmitted or processed.

78. What is Variety?

Answer: Different forms and structures of data.

79. What is Veracity?

Answer: The quality, reliability and trustworthiness of data.

80. What is a Data Lake?

Answer: A centralized repository capable of storing large volumes of raw structured, semi-structured and unstructured data.

81. Data Warehouse vs Data Lake?

Answer: Warehouses generally store curated structured data for analytics, while data lakes can store raw data in multiple formats.

82. What is NoSQL?

Answer: A category of databases designed for flexible data models and scalable distributed workloads.

83. Give examples of NoSQL database models.

Answer: Key-value, document, wide-column and graph databases.

SECTION 8: AI, ML, DL AND GENAI

These topics are explicitly part of the curriculum.

84. What is Artificial Intelligence?

Answer: The broader field of creating systems capable of performing tasks associated with human intelligence.

85. What is Machine Learning?

Answer: A subset of AI where systems learn patterns from data to make predictions or decisions.

86. What is Deep Learning?

Answer: A subset of machine learning based primarily on neural networks with multiple layers.

87. AI vs ML?

Answer: AI is the broader field; ML is one approach used to build AI systems.

88. Supervised vs unsupervised learning?

Answer:

- Supervised learning uses labeled data.
- Unsupervised learning works with unlabeled data.

89. Classification vs regression?

Answer:

- Classification predicts categories.
- Regression predicts continuous numerical values.

90. What is overfitting?

Answer: When a model learns the training data too closely and performs poorly on unseen data.

91. What is underfitting?

Answer: When a model is too simple to capture the underlying patterns in the data.

92. What is a training dataset?

Answer: Data used to train a machine-learning model.

93. What is a validation dataset?

Answer: Data used during development to tune model parameters and evaluate intermediate performance.

94. What is a test dataset?

Answer: Data used to evaluate final model performance on unseen examples.

95. What is Generative AI?

Answer: AI that generates new content such as text, images, audio, video or code.

96. What is an LLM?

Answer: A Large Language Model is a model trained on large amounts of text to understand and generate language.

SECTION 9: DEEP LEARNING, CNN, RNN AND TRANSFORMERS

The curriculum specifically includes RNNs, Transformers and CNNs.

97. What is a neural network?

Answer: A computational model composed of interconnected neurons arranged in layers.

98. What is an activation function?

Answer: A function that introduces non-linearity into a neural network.

99. Give examples of activation functions.

Answer: ReLU, sigmoid and tanh.

100. What is a CNN?

Answer: A Convolutional Neural Network is particularly effective for spatial data such as images.

101. What does a convolution layer do?

Answer: It applies learnable filters to extract local patterns and features from input data.

102. What is pooling?

Answer: A dimensionality-reduction operation that summarizes feature maps.

103. What is an RNN?

Answer: A Recurrent Neural Network processes sequential data while maintaining information from previous steps.

104. What problem can traditional RNNs face?

Answer: Vanishing and exploding gradient problems, particularly for long sequences.

105. What is an LSTM?

Answer: Long Short-Term Memory is an RNN architecture designed to better preserve information over longer sequences.

106. What is a Transformer?

Answer: A neural-network architecture based primarily on attention mechanisms and designed to efficiently process sequential or tokenized data.

107. What is self-attention?

Answer: A mechanism allowing each token to determine how much other tokens in the sequence should influence its representation.

108. Why are Transformers important for GenAI?

Answer: They efficiently model relationships between tokens and can be scaled to very large models.

SECTION 10: REINFORCEMENT LEARNING

109. What is Reinforcement Learning?

Answer: A learning paradigm where an agent learns by interacting with an environment and receiving rewards or penalties.

110. What is an agent?

Answer: The entity that takes actions in an environment.

111. What is a state?

Answer: A representation of the current situation of the environment.

112. What is an action?

Answer: A decision or operation taken by the agent.

113. What is a reward?

Answer: Feedback indicating how desirable an action or outcome was.

114. What is a policy?

Answer: A strategy mapping states to actions.

SECTION 11: AGENTIC AI AND RAG

The syllabus specifically includes Agentic AI Fundamentals and later Databricks Agentic Application topics such as Vector Search, RAG, Mosaic AI Agent Framework and Unity Catalog tool functions.

115. What is Agentic AI?

Answer: AI systems capable of planning, reasoning, selecting tools and executing actions toward a goal.

116. Traditional chatbot vs AI agent?

Answer: A traditional chatbot primarily responds to user input, while an agent can reason through tasks, use tools and perform multi-step actions.

117. What is RAG?

Answer: Retrieval-Augmented Generation combines information retrieval with generative AI so the model can use relevant external information when generating responses.

118. What are the major steps in a RAG pipeline?

Answer:

1. Ingest documents.
2. Chunk documents.
3. Generate embeddings.
4. Store embeddings.
5. Retrieve relevant chunks.
6. Provide retrieved context to the LLM.
7. Generate the response.

119. What is an embedding?

Answer: A numerical vector representation of data such as text that captures semantic information.

120. What is vector search?

Answer: Searching for vectors that are semantically similar to a query vector.

121. Why use RAG instead of relying only on an LLM?

Answer: RAG can provide external, domain-specific or updated information without requiring the model itself to be retrained.

122. What is hallucination in GenAI?

Answer: When a model generates information that is unsupported, incorrect or fabricated.

123. How can RAG reduce hallucinations?

Answer: By retrieving relevant source information and providing it as context to the generation model.

124. What is tool calling?

Answer: Allowing an AI model or agent to invoke external functions, APIs or services.

125. What is an agentic workflow?

Answer: A workflow where an AI system can determine steps, invoke tools, observe results and continue toward a goal.

SECTION 12: DATA WAREHOUSING

The syllabus includes DW characteristics, OLTP/OLAP, schemas, ETL/ELT, architecture and data modeling.

126. What is a Data Warehouse?

Answer: A centralized analytical data store designed for reporting and decision-making.

127. What is OLTP?

Answer: Online Transaction Processing systems handle operational transactions such as inserts, updates and deletes.

128. What is OLAP?

Answer: Online Analytical Processing systems are optimized for analytical queries and aggregations.

129. OLTP vs OLAP?

Answer:

OLTP	OLAP
Transactions	Analytics
Frequent writes	Mostly reads
Highly normalized	Often dimensional/denormalized
Current operational data	Historical/analytical data

130. What is a fact table?

Answer: A table containing measurable business events or metrics.

131. What is a dimension table?

Answer: A table containing descriptive attributes used to analyze facts.

132. What is a star schema?

Answer: A dimensional model with a central fact table connected directly to dimension tables.

133. What is a snowflake schema?

Answer: A dimensional schema where dimension tables are further normalized into related tables.

134. Star schema vs snowflake schema?

Answer: Star schemas are generally simpler and more denormalized, while snowflake schemas normalize dimensions more extensively.

135. What is ETL?

Answer: Extract, Transform and Load.

136. What is ELT?

Answer: Extract, Load and Transform.

137. ETL vs ELT?

Answer: ETL transforms data before loading it into the target, whereas ELT loads data first and performs transformations within the target platform.

SECTION 13: DATA MODELING

138. What is data modeling?

Answer: The process of designing how data entities, attributes and relationships are represented.

139. What is a conceptual data model?

Answer: A high-level representation of major entities and their relationships.

140. What is a logical data model?

Answer: A detailed representation of entities, attributes and relationships independent of a particular database implementation.

141. What is a physical data model?

Answer: The database-specific implementation of the logical model.

142. What is a one-to-one relationship?

Answer: One record in one entity corresponds to one record in another entity.

143. What is a one-to-many relationship?

Answer: One record can correspond to multiple records in another entity.

144. What is a many-to-many relationship?

Answer: Multiple records in one entity can correspond to multiple records in another entity.

SECTION 14: DATA INTEGRATION AND TALEND

The syllabus includes Talend components and job creation/deployment.

145. What is data integration?

Answer: Combining data from multiple sources into a unified system or pipeline.

146. What is Talend?

Answer: A data integration platform used to build data pipelines and integration workflows.

147. What is a Talend job?

Answer: A workflow containing components that perform data integration or processing tasks.

148. What is a source in a data pipeline?

Answer: The system from which data originates.

149. What is a target?

Answer: The destination to which processed data is written.

150. What is an ETL pipeline?

Answer: A sequence of operations that extracts data, transforms it and loads it into a target system.

SECTION 15: APACHE SPARK

This is one of the major sections of the uploaded syllabus. It covers Spark architecture, services, ecosystem, distributed processing, Python and RDDs.

151. What is Apache Spark?

Answer: An open-source distributed computing engine designed for large-scale data processing.

152. Why is Spark used for Big Data?

Answer: It distributes computation across multiple machines and supports scalable processing of large datasets.

153. What is a Spark cluster?

Answer: A group of machines working together to execute Spark workloads.

154. What is the Spark driver?

Answer: The process responsible for coordinating a Spark application.

155. What is an executor?

Answer: A process running on worker nodes that executes tasks and stores data for the application.

156. What is a worker node?

Answer: A machine that provides computational resources for Spark executors.

157. What is a Spark job?

Answer: A collection of computations triggered by an action.

158. What is a Spark stage?

Answer: A group of tasks that can execute without requiring a shuffle boundary.

159. What is a Spark task?

Answer: The smallest unit of work sent to an executor for execution.

160. What is lazy evaluation?

Answer: Spark delays execution of transformations until an action requires the result.

161. Why does Spark use lazy evaluation?

Answer: It allows Spark to optimize the execution plan before actually processing the data.

162. What is a transformation?

Answer: An operation that creates a new dataset from an existing dataset.

163. Give examples of Spark transformations.

Answer: map, filter, select, groupBy and join.

164. What is an action?

Answer: An operation that triggers execution and produces a result.

165. Give examples of Spark actions.

Answer: count(), collect(), show(), first() and write.

SECTION 16: RDD

RDD concepts are explicitly included in the syllabus.

166. What is an RDD?

Answer: Resilient Distributed Dataset is a distributed, fault-tolerant collection of data that can be processed in parallel.

167. What does “Resilient” mean in RDD?

Answer: RDDs can recover lost partitions using lineage.

168. What does “Distributed” mean?

Answer: Data can be partitioned and distributed across multiple machines.

169. What is RDD lineage?

Answer: The record of transformations used to construct an RDD.

170. Why is lineage important?

Answer: Spark can recompute lost partitions using the transformation history.

171. RDD vs DataFrame?

Answer: RDDs provide a lower-level distributed data abstraction, while DataFrames provide structured data with query optimization.

SECTION 17: PYSPARK DATAFRAMES AND SQL

The syllabus explicitly includes PySpark DataFrames, PySpark SQL and streaming.

172. What is PySpark?

Answer: The Python API for Apache Spark.

173. What is a PySpark DataFrame?

Answer: A distributed collection of data organized into named columns.

174. Why are DataFrames generally preferred over RDDs for structured data?

Answer: They provide schema information and allow Spark's optimizer to optimize execution.

175. What is Spark SQL?

Answer: Spark's module for processing structured data using SQL and DataFrame APIs.

176. What is a schema?

Answer: The structure of a dataset, including column names and data types.

177. What does `select()` do?

Answer: Selects specified columns or expressions.

178. What does `filter()` do?

Answer: Returns rows satisfying a specified condition.

179. What does `withColumn()` do?

Answer: Creates or replaces a DataFrame column.

180. What does `groupBy()` do?

Answer: Groups records based on one or more columns for aggregation.

181. What is a Spark shuffle?

Answer: Redistribution of data across partitions, often required by operations such as joins and aggregations.

182. Why can shuffles be expensive?

Answer: They involve data movement across the cluster, which can cause network, disk and serialization overhead.

SECTION 18: PYSPARK PERFORMANCE

183. What is partitioning?

Answer: Dividing distributed data into partitions that can be processed in parallel.

184. What is repartition?

Answer: An operation that reshuffles data to create a specified number of partitions.

185. What is coalesce?

Answer: An operation generally used to reduce the number of partitions with less or no full shuffle in common cases.

186. What is caching?

Answer: Keeping computed data in memory or another storage level for reuse.

187. When should you cache a DataFrame?

Answer: When the same computed dataset is reused multiple times and caching provides a performance benefit.

188. What is a broadcast join?

Answer: A join where a small dataset is broadcast to executors so that a large shuffle can potentially be avoided.

189. When is broadcast join useful?

Answer: When one side of the join is sufficiently small to be efficiently broadcast.

190. What is data skew?

Answer: Uneven distribution of data across partitions.

191. Why is data skew problematic?

Answer: Some tasks receive significantly more data and become stragglers, slowing the entire job.

SECTION 19: SPARK STREAMING

192. What is stream processing?

Answer: Processing data continuously as it arrives rather than waiting for a complete batch.

193. Batch processing vs streaming?

Answer: Batch processes bounded datasets, while streaming processes continuously arriving data.

194. What is a streaming source?

Answer: A system from which streaming data is consumed.

195. What is a streaming sink?

Answer: The destination where streaming results are written.

196. What is a checkpoint in Spark Streaming?

Answer: Persistent state information used to recover streaming applications after failures.

SECTION 20: SYNTHETIC DATA

The syllabus includes synthetic data, comparisons with anonymized/real data, dataset creation and validation.

197. What is synthetic data?

Answer: Artificially generated data designed to resemble characteristics of real data.

198. Synthetic vs real data?

Answer: Real data originates from actual observations or events; synthetic data is artificially generated.

199. Synthetic vs anonymized data?

Answer: Synthetic data is generated artificially, while anonymized data originates from real records with identifying information removed or transformed.

200. Why use synthetic data?

Answer: Testing, development, privacy-sensitive scenarios, simulation and generating data when real datasets are unavailable.

201. What is synthetic data utility?

Answer: How well synthetic data preserves useful characteristics and relationships of the original data.

202. What is data quality validation?

Answer: Checking properties such as completeness, consistency, accuracy, validity and distribution characteristics.

SECTION 21: DATABRICKS

The syllabus has a dedicated Databricks refresher covering the platform, Lakehouse, Delta Lake, clusters, DBFS, interfaces and ingestion.

203. What is Databricks?

Answer: A cloud-based data and AI platform built around technologies including Apache Spark and Delta Lake.

204. What is the Lakehouse architecture?

Answer: An architecture combining characteristics of data lakes and data warehouses.

205. Why is Lakehouse architecture useful?

Answer: It enables scalable storage of diverse data while providing structured data management and analytics capabilities.

206. What is Delta Lake?

Answer: An open-source storage layer that adds transactional and reliability capabilities to data lakes.

207. What is ACID?

Answer:

- Atomicity
- Consistency
- Isolation
- Durability

208. Why is ACID important in data systems?

Answer: It helps maintain reliable and consistent transactions.

209. What is a Databricks cluster?

Answer: A collection of compute resources used to execute Databricks workloads.

210. What is DBFS?

Answer: Databricks File System is a storage abstraction historically used to interact with data in Databricks environments.

211. What is the purpose of Databricks notebooks?

Answer: To interactively develop, execute and document code and data workflows.

212. What is Databricks Assistant?

Answer: An AI-assisted capability within Databricks intended to help users with tasks such as coding and data-related workflows.

SECTION 22: DELTA LAKE

213. What is a Delta table?

Answer: A table whose underlying storage uses Delta Lake.

214. What is the transaction log in Delta Lake?

Answer: A log that records changes and metadata associated with table transactions.

215. What is time travel?

Answer: The ability to query previous versions of a Delta table.

216. Why is time travel useful?

Answer: It can support auditing, debugging, historical analysis and recovery from accidental changes.

217. What is schema enforcement?

Answer: Preventing incompatible data from being written according to the table schema.

218. What is schema evolution?

Answer: Allowing controlled changes to a table's schema as new data arrives.

SECTION 23: DATABRICKS VECTOR SEARCH AND RAG

The syllabus specifically lists Vector Search and RAG under Advanced Databricks.

219. What is Databricks Vector Search used for?

Answer: Searching vector representations of data to retrieve semantically relevant information.

220. What is an embedding model?

Answer: A model that converts data such as text into numerical vectors.

221. What happens during document chunking?

Answer: Large documents are divided into smaller pieces suitable for embedding and retrieval.

222. Why shouldn't documents always be embedded as one huge chunk?

Answer: Smaller relevant chunks generally allow more precise retrieval and provide focused context to the LLM.

223. What is semantic similarity?

Answer: A measure of how closely two pieces of content are related in meaning.

224. What happens after vector retrieval in RAG?

Answer: Relevant retrieved content is provided to the generative model as context for generating an answer.

SECTION 24: UNITY CATALOG

225. What is Unity Catalog?

Answer: A Databricks governance solution for managing and governing data and AI assets.

226. Why is data governance important?

Answer: It helps control access, discover assets, maintain lineage and enforce organizational policies.

227. What is data lineage?

Answer: Information about where data originates, how it changes and where it flows.

228. What are Unity Catalog functions used for in agentic applications?

Answer: They can expose governed functions/tools that an AI agent can invoke as part of its workflow.

SECTION 25: DEVOPS AND DATAOPS

The syllabus covers DevOps and DataOps.

229. What is DevOps?

Answer: A set of practices combining development and operations to deliver software faster and more reliably.

230. What is DataOps?

Answer: A set of practices applying automation, collaboration and quality controls to data engineering and analytics workflows.

231. CI vs CD?

Answer:

- **CI:** Continuous Integration, frequently integrating and validating code changes.
- **CD:** Continuous Delivery/Deployment, automating delivery or deployment of software.

232. What is automation in DevOps?

Answer: Using tools and pipelines to automatically perform activities such as testing, building and deployment.

SECTION 26: GIT AND GITHUB

The curriculum includes GitHub fundamentals and Git fundamentals.

233. What is Git?

Answer: A distributed version-control system.

234. What is GitHub?

Answer: A platform for hosting Git repositories and collaborating on software projects.

235. What is a repository?

Answer: A location containing a project's files and version history.

236. What is a commit?

Answer: A recorded snapshot of changes in a Git repository.

237. What is a branch?

Answer: An independent line of development.

238. What is merging?

Answer: Combining changes from one branch into another.

239. What is a pull request?

Answer: A request to review and merge changes from one branch into another.

240. What is a merge conflict?

Answer: A situation where Git cannot automatically reconcile conflicting changes.

241. What is .gitignore?

Answer: A file specifying files and directories Git should not track.

SECTION 27: GITHUB ACTIONS

The syllabus explicitly includes GitHub Actions.

242. What is GitHub Actions?

Answer: A GitHub automation and CI/CD platform.

243. What is a workflow?

Answer: An automated process defined using a workflow configuration.

244. What triggers a GitHub Actions workflow?

Answer: Events such as pushes, pull requests, schedules or manual triggers.

245. What is a job?

Answer: A group of steps executed on a runner.

246. What is a runner?

Answer: A machine or environment that executes GitHub Actions jobs.

247. Why use GitHub Actions?

Answer: To automate testing, building, deployment and other development workflows.

SECTION 28: DOCKER

The syllabus explicitly covers Docker fundamentals.

248. What is Docker?

Answer: A platform for packaging and running applications in containers.

249. What is a container?

Answer: An isolated execution environment containing an application and its dependencies.

250. What is a Docker image?

Answer: A read-only template used to create containers.

251. Image vs container?

Answer: An image is the packaged template; a container is a running instance of an image.

252. What is a Dockerfile?

Answer: A file containing instructions for building a Docker image.

253. Why use containers?

Answer: They provide consistent application environments, portability and isolation.

254. Containers vs virtual machines?

Answer: Containers share the host OS kernel, while VMs generally include a complete guest operating system.

SECTION 29: KUBERNETES

255. What is Kubernetes?

Answer: An open-source platform for orchestrating containerized applications.

256. Why use Kubernetes?

Answer: To automate deployment, scaling, networking, service discovery and management of containers.

257. What is a Pod?

Answer: The smallest deployable unit in Kubernetes, usually containing one or more containers.

258. What is a Deployment?

Answer: A Kubernetes object used to manage replicated application Pods.

259. What is a Kubernetes Service?

Answer: A stable networking abstraction that exposes applications running in Pods.

260. What is horizontal scaling?

Answer: Increasing or decreasing the number of application instances.

261. What is a Kubernetes cluster?

Answer: A collection of nodes managed by Kubernetes.

SECTION 30: MICROSERVICES

The syllabus includes Microservices Architecture and deployment using AWS and Kubernetes.

262. What is microservices architecture?

Answer: An architecture where an application is divided into independently deployable services.

263. Monolith vs microservices?

Answer: A monolith generally deploys an application as one unit, while microservices divide it into independently deployable services.

264. Advantage of microservices?

Answer: Independent deployment, scaling and development of individual services.

265. What is service discovery?

Answer: The mechanism by which services locate other services dynamically.

266. Why is Kubernetes commonly used with microservices?

Answer: It provides orchestration, service discovery, scaling, deployment and container management.

HIGH-VALUE SCENARIO QUESTIONS

These are the types of questions I would **especially practice for an internal technical assessment**.

267. Your Spark job is slow because one partition contains 90% of the data. What is the likely problem?

Answer: Data skew.

268. A Spark DataFrame is used five times in the same application. What optimization might help?

Answer: Caching/persisting the DataFrame, provided the memory and workload justify it.

269. You need to join a 1 TB dataset with a 5 MB lookup table. What optimization might be appropriate?

Answer: A broadcast join may be appropriate because the lookup table is very small.

270. A Spark transformation has been defined but no action has been called. Has Spark necessarily executed it?

Answer: No. Spark uses lazy evaluation, so execution generally begins when an action is triggered.

271. You need to store millions of images in AWS. Which service is appropriate?

Answer: Amazon S3.

272. You need a relational database for an application but don't want to manage the database infrastructure yourself. Which AWS service is suitable?

Answer: Amazon RDS.

273. Your application suddenly receives 10 times its normal traffic. What cloud capability helps automatically adjust resources?

Answer: Auto Scaling/elasticity.

274. You have sensitive backend servers that should not be directly accessible from the internet. Where would you normally place them?

Answer: A private subnet.

275. You need a serverless function that executes when a file is uploaded to S3. What would you use?

Answer: AWS Lambda triggered by an S3 event.

276. You need to query structured data using SQL in Spark. What can you use?

Answer: Spark SQL or the DataFrame API.

277. You need to retrieve documents based on semantic meaning rather than exact keywords. What technology is appropriate?

Answer: Vector search using embeddings.

278. An LLM needs to answer questions using a company's private documents. What architecture is appropriate?

Answer: RAG.

279. Your documents change frequently and you don't want to retrain the LLM every time. What approach can help?

Answer: RAG, because the knowledge can be updated in the retrieval layer.

280. An AI system needs to query a database and then use the result to answer the user. What capability does this require?

Answer: Tool/function calling within an agentic workflow.

281. You need historical versions of a dataset for auditing. Which Delta Lake capability is useful?

Answer: Time travel.

282. Multiple applications need independently deployable components. What architecture fits this requirement?

Answer: Microservices architecture.

283. You want to package an application together with its dependencies so it runs consistently across environments. What should you use?

Answer: Docker containers.

284. You have hundreds of containers and need automated scaling and orchestration. What technology is appropriate?

Answer: Kubernetes.

285. Developers push code to GitHub and you want automated tests to run automatically. What can you use?

Answer: GitHub Actions.

RAPID-FIRE QUESTIONS

These are good for last-minute revision.

286. What does AWS stand for?

Answer: Amazon Web Services.

287. What does EC2 stand for?

Answer: Elastic Compute Cloud.

288. What does S3 stand for?

Answer: Simple Storage Service.

289. What does RDS stand for?

Answer: Relational Database Service.

290. What does VPC stand for?

Answer: Virtual Private Cloud.

291. What does IAM stand for?

Answer: Identity and Access Management.

292. What does SQL stand for?

Answer: Structured Query Language.

293. What does ETL stand for?

Answer: Extract, Transform, Load.

294. What does ELT stand for?

Answer: Extract, Load, Transform.

295. What does OLTP stand for?

Answer: Online Transaction Processing.

296. What does OLAP stand for?

Answer: Online Analytical Processing.

297. What does RDD stand for?

Answer: Resilient Distributed Dataset.

298. What does RAG stand for?

Answer: Retrieval-Augmented Generation.

299. What does LLM stand for?

Answer: Large Language Model.

300. What does CNN stand for?

Answer: Convolutional Neural Network.

301. What does RNN stand for?

Answer: Recurrent Neural Network.

302. What does NoSQL mean?

Answer: Non-relational database technologies designed for flexible and scalable data models.

303. What does CI mean?

Answer: Continuous Integration.

304. What does CD mean?

Answer: Continuous Delivery or Continuous Deployment, depending on context.

305. What does API stand for?

Answer: Application Programming Interface.

306. What does DBFS stand for?

Answer: Databricks File System.

307. What is Delta Lake?

Answer: A storage layer providing reliability and transactional capabilities on data lakes.

308. What is a Data Lake?

Answer: A repository capable of storing large volumes of raw data in different formats.

309. What is a Data Warehouse?

Answer: A system optimized for analytical processing and reporting.

310. What is Docker?

Answer: A platform for containerizing and running applications.

311. What is Kubernetes?

Answer: A container orchestration platform.

312. What is Git?

Answer: A distributed version-control system.

313. What is GitHub?

Answer: A platform for hosting and collaborating on Git repositories.

The areas I would prioritize

Based on the **structure of your uploaded syllabus**, I would spend particularly serious preparation time on:

Tier 1

1. **PySpark**
2. **Databricks**
3. **SQL**
4. **AWS**
5. **Data Warehousing/Data Modeling**

Tier 2

6. Python
7. Big Data
8. RAG/GenAI/Agentic AI
9. Git/GitHub/GitHub Actions
10. Docker/Kubernetes

Tier 3

11. Talend
12. Synthetic Data
13. Deep Learning/RNN/CNN/Transformers
14. Reinforcement Learning
15. Microservices

That weighting follows the syllabus structure, which gives substantial dedicated coverage to Spark/PySpark and Databricks, including advanced agentic application material.

Most important next step

For your **Accenture internal exam**, I would not stop at basic Q&A. The next preparation set should be a **hard 200-question mock bank**, especially around:

- PySpark output prediction
- Spark transformations vs actions
- partitioning and shuffles
- joins and broadcast joins

- DataFrame operations
- SQL query problems
- AWS service selection scenarios
- Delta Lake
- Databricks architecture
- RAG and Vector Search
- Agentic AI
- Git/GitHub
- Docker/Kubernetes
- ETL/ELT
- Data warehouse schemas

Those questions are much closer to the kind of thing that can distinguish someone who has memorized definitions from someone who actually understands the technologies.